

Sizing Up Washington: Are Women in Washington State More Petite Than the U.S. Average?

CSE 163 Final Project Proposal | Elly Lee | Student #2163622

Research Questions

- RQ1:** How do the average body measurements (arm length, inseam, waist circumference, shoulder width) of Washington state women compare to the U.S. national average?
- RQ2:** How do Washington state women's body measurements align with Ann Taylor's published size guide, and which product categories show the greatest sizing mismatch?
- RQ3:** Is Washington state's female demographic composition distinct enough from the U.S. average to warrant region-specific sizing considerations for retailers?
- RQ4:** Are the body measurement differences between Washington state women and the U.S. average statistically significant enough to justify region-specific sizing adjustments?

Motivation

The U.S. women's apparel industry is largely calibrated to the body proportions of white American women. Washington state has one of the highest concentrations of Asian-American residents of any U.S. state — a population whose women tend to have smaller and shorter body proportions on average. However, women of other ethnicities in Washington may also be more petite, and this project does not assume Asian-American women are the sole driver of any observed difference.

As a sales associate at the Ann Taylor location in Bellevue, WA — the only Ann Taylor store in the entire state — I have observed that **"too big/small"** is consistently the most frequently selected return reason in our store's return system. Because this single store serves all of Washington state, a sizing mismatch here affects Ann Taylor's entire WA customer base. Notably, Ann Taylor already carries a Petite fit category — yet returns citing size mismatch persist even on petite products, suggesting that the existing petite line is still not adequately proportioned for Washington state women. This project aims to quantify that gap with data, and to provide a measurable foundation for more targeted sizing decisions in the Pacific Northwest market.

Datasets

Dataset	Source URL	Used For
ANSUR II (Women) 2012	openlab.psu.edu/ansur2	Structural measurements: arm length, inseam, shoulder width
NHANES Body Measurements 2015–2018	openlab.psu.edu/nhanes	Circumference measurements: waist, hip (pre-merged CSV, women 20+)

ACS PUMS 2020–2024	census.gov/programs-surveys/acs/microdata/access.html	WA state racial/ethnic demographic weights
Ann Taylor Size Guide (Classic + Petite)	anntaylor.com — accessible via 'Size Chart' button on any product page (modal popup)	Reference sizing standard for Classic and Petite categories; recorded May 2026

Challenge Goals

1. Multiple Datasets

Three datasets are combined to answer the research questions. ANSUR II and NHANES provide body measurements broken down by race/ethnicity at the national level. ACS PUMS provides Washington state's female demographic composition. These are joined by mapping race/ethnicity codes across datasets and applying Census demographic weights to produce a Washington state-representative estimate. At least two RQs (RQ1, RQ2) involve all three datasets, and at least one explicit merge/join is performed.

2. Result Validity

RQ4 applies independent samples t-tests (`scipy.stats.ttest_ind`) to verify that observed measurement differences between Washington state women and the U.S. average are statistically significant ($\alpha = 0.05$). Cohen's d effect sizes are computed alongside p-values to confirm practical as well as statistical significance. Results are clearly interpreted in the context of the research questions.

Methodology

RQ1 — Measurement Comparison

Load ANSUR II and NHANES; filter to female respondents. The Penn State NHANES CSV (NHANES15-18_womenAge20YearsAndOver.csv) is already filtered to women aged 20+ with demographics and body measures combined — no merge required. Key columns: RIDRETH3 (race/ethnicity), BMXWAIST (waist, mm), WTMEC2YR (sample weight). Load ACS PUMS WA and national files; compute demographic weight vectors using PWGTP, RAC1P, HISP. Compute per-group measurement means grouped by race/ethnicity in both datasets. Apply weights to produce Washington-weighted and national average estimates. Note: ANSUR II uses mm; Penn State NHANES also uses mm — convert all to cm before combining. Visualize with grouped bar charts.

RQ2 — Ann Taylor Size Alignment

Manually transcribe Ann Taylor's Classic and Petite size charts — accessed via the 'Size Chart' modal on any product page at anntaylor.com — into two separate pandas DataFrames; convert inches to cm. Compare Washington-weighted averages from RQ1 against size ranges for tops, pants, and dresses for both Classic and Petite fits. Identify the best-fitting size per category in each fit chart; compute mismatch metric (difference from size midpoint in cm). Since Ann Taylor already offers a Petite line yet sizing mismatches persist at our Bellevue store, this dual comparison directly tests whether the existing Petite chart is adequate for Washington state women or whether a further regional adjustment is warranted. Visualize with annotated range plots per product category showing both Classic and Petite ranges alongside the Washington state average.

RQ3 — Demographic Divergence

Using ACS PUMS weight vectors from RQ1, compute per-group proportion differences between Washington state and the U.S. national female population. Compute Total Variation Distance (TVD) as a summary divergence metric. Visualize with stacked bar chart (WA vs. U.S.) and divergence bar chart sorted by group-level difference.

RQ4 — Statistical Significance

Resample ANSUR II and NHANES rows proportionally to demographic weights to construct individual-level Washington-representative and national samples. Run `scipy.stats.ttest_ind` for each measurement (sleeveoutseam, crotchheight, biacromialbreadth, waistcircumference from ANSUR II; BMXARML, BMXLEG, BMXWAIST, BMXHIP from NHANES). Compute Cohen's d for significant results. Visualize with forest plot and p-value heatmap.

Work Plan

Task	Description	Est. Hours
1	Data acquisition, environment setup, Ann Taylor size guide transcription	3
2	Data cleaning, filtering, race/ethnicity code harmonization across datasets	5
3	RQ1 + RQ3: demographic weights, measurement comparisons, visualizations	6
4	RQ2: Ann Taylor size alignment analysis and range plot visualizations	4
5	RQ4: resampling, t-tests, Cohen's d, forest plot and heatmap	5
6	Report writing, code cleanup, testing module, final review	4
	Total	27

Workflow note: Tasks 1 and 2 must be completed before any analysis begins, as the race/ethnicity harmonization in Task 2 underpins all subsequent computations. If Task 2 runs over schedule, RQ3's TVD computation is the safest place to simplify. Code will be prototyped in Jupyter Notebooks before being transferred into a final runnable Python script as required by the project spec.